WORKING GROUP KICKOFF · NOVEMBER 2026

Agent School: what's coming

A new 4xx platform-layer series beside the shipped courses: skills as a decoupled component, RAG placed by rule, a telecom-tuned model track, plus a syllabus that teaches the blueprint hands-on.

open-experiments/agent-school · Red Hat · AT&T · Microsoft Azure · AMD

Where we are

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courses live on the Rome and Venice reference clusters, each a Telco-AIX experiment rebuilt as a governed workload

5 + 5

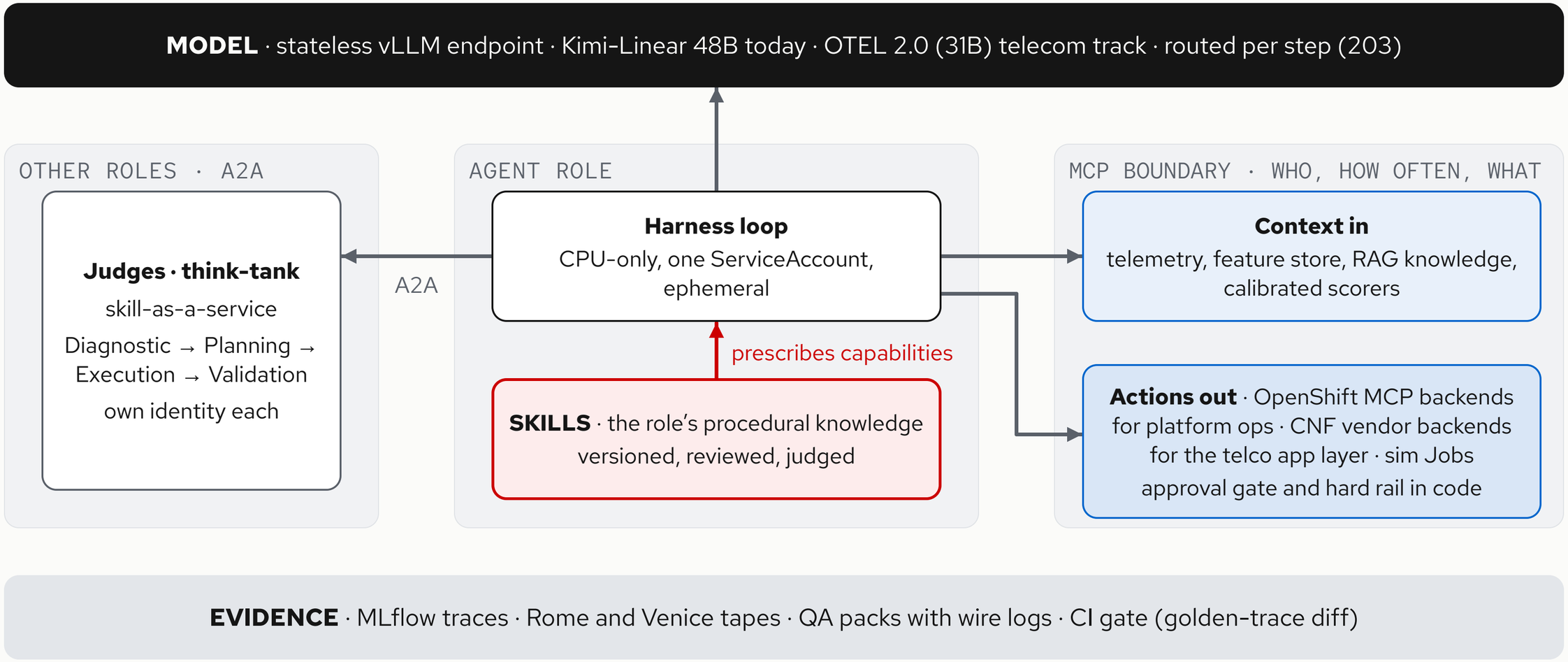
narrated walkthroughs and interactive labs that replay captured cluster tapes; nothing a learner sees is synthesized

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blueprint Table 1 rows covered (tool governance strong), 3 partial, 3 skipped; claims carry captured evidence

Table 1 status	Rows
Covered	Harness (custom loops, LangGraph, Llama Stack), model (vLLM), model workers, skills as MCP tools, orchestration (A2A), tool governance (Kuadrant AuthPolicy and RateLimit)
Partial	Agent blueprint and GitOps, agentic loop runtime, AI gateway and MaaS
Skipped	Sandbox runtime (102 authored, unproven), semantic routing, inference routing, workload identity

Skill, capability, role: three things



401 Skills and capabilities

Skill catalog

- One SKILL.md per role
- Frontmatter: role, risk class, prescribed capabilities, offline episode
- Governed as content: review, version, provenance, a judge
- Portable across harnesses; the OpenClaw track regains its procedure

Capability catalog

- A manifest per MCP server: backend pattern 1/2/3, I/O schema, risk class, allowed identities, rate class, data class
- Generates the registry, per-agent allow-lists, Kuadrant AuthPolicy and RateLimit
- One source for the six copies of 301's playbook catalog

Selection, two layers

- Procedure per problem class: role level, mostly static, routed by class as the library grows
- Capability per step: the skill prescribes it; cheap reads first, actuation last
- Which skill, which capability, and why: recorded in the trace

THE FIRST PLATFORM-LAYER COURSE, PROVEN ON THE SHIPPED ONES

101, 301 and 302 consume the registry exactly as shipped. Their deviations today (procedures in prompts and Python, 202 and 301 reads outside MCP, six copies of one catalog) are recorded in the syllabus, with no patches to their code, tapes or videos.

402 Knowledge: RAG placed by rule

SKILL

When the role owns the procedure. A runbook used as the NOC assistant's own triage procedure is a skill.

CONTEXT THROUGH MCP (RAG)

For reference knowledge the domain owns: vendor docs, specs, past RCAs, alarm dictionaries, change records. Large, citable, kept inside the trust boundary.

STRUCTURED TOOLS

For state: KPIs, alerts, feature store, registry. Never a text index over telemetry rows.

What 402 builds beside the shipped courses

Beside	Retrieval, placed by the rule
201	A knowledge service (runbooks, past RCAs, alarm dictionary) stood up next to 201 and compared with its KPI-rows-as-text corpus; 201 stays as shipped
401	Retrieval over the capability inventory, so selection scales past a flat tool list
301	Case memory over past loop episodes and gate decisions (feeds issue #19); topology retrieval for the blast-radius context the judge lacks; 301 untouched
Platform	Hand-rolled FastAPI + TF-IDF, or Llama Stack vector_io as the platform-provided candidate

403 Domain model track: OTEL 2.0

The model

- Gemma 4 31B-IT, post-trained (OSFT) on about 440B telecom tokens: 3GPP, ETSI, ITU, GSMA, CAMARA, O-RAN, TM Forum, plus the Telco Corpus
- Apache-2.0; vLLM 0.19+ in bf16; TP=1 on 192 to 288 GB cards, TP=2 on 80 GB
- Top of the Open Telco AI leaderboard; released July 24, 2026

What its card says

- The training mix has no telecom-specific tool-calling, MCP or instruction-following data
- Not benchmarked for closed-loop network tasks; tool readiness unproven
- Recommends retrieval, source attribution and human review for high-impact use

What 403 measures, then tunes

- Tool-calling on the 101 and 201 tapes, with wire logs as evidence
- Judge groundedness with the 302 suite; closed-loop behavior in 301 (co-decision, gate, rollback)
- Cost per resolved episode as the small tier in 201 routing and course 203
- Then SDG Hub turns the 401 catalogs and the tapes into the telecom tool-calling and MCP data the card lacks, and OSFT folds it in without forgetting

SERVING HOME OPTIONS

AMD MI300X or MI355X at TP=1 · Azure GPU · Venice's 96 GB card at TP=1 with a short context · a MaaS endpoint. The courses swap models by config: the endpoint contract stays the same.

Syllabus upgrade: blueprint-wise and hands-on

BLUEPRINT-WISE

One SYLLABUS.md that ties the courses to the article

- Component × course matrix: model, harness, skills, capabilities, governance, state, observability, evaluation, data pipeline
- Learning objectives per course and a prerequisites graph
- Decision guides: which harness, which capability pattern (1 thin client, 2 service, 3 batch Job), when RAG, a classic model as a tool, when a judge, which rail forces which safe state
- Glossary: skill and capability as the 4xx series defines them; the shipped courses keep their wording, with their deviations recorded here instead of patched

HANDS-ON

A fourth lab tier: modify

- Today: replay in the portal, run offline on a laptop, deploy on a cluster
- Add exercises with a checker: add and register a skill; call it from the wrong identity and capture the 403; swap the retriever; tighten a gate and capture the audited override
- The CI evidence gate (discussion #21) is the checker, so exercises and CI are one investment

Roadmap: gated steps, scope, partner lanes

STEP 1

Design note

4xx definition, contracts, RAG rules, syllabus skeleton

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STEP 2

401 Skills and capabilities

catalogs by tier, registry, policies, onboarding contract

→

STEP 3

402 Knowledge

knowledge service beside 201, case memory and topology beside 301

→

STEP 4

403 Domain model track

OTEL 2.0 measured on the tapes, then SDG Hub and OSFT

→

STEP 5

Syllabus and modify labs

matrix, guides, glossary, exercises with a checker

SCOPE OF THIS CYCLE

In: 401, then 402, then 403, plus the syllabus. 401 goes first because 402 retrieves over its inventory and 403's tuning data comes from its catalogs.

Out, unchanged on the public roadmap: 102 sandbox, 203 routing, 303 identity and sandbox, 304 agent SecOps.

The shipped courses stay as they are, and each step is reviewed before the next starts.

CANDIDATE LANES, FOR THE WG TO CONFIRM

Party	Lane	Owner, date
Red Hat	Platform and the 4xx builds: OpenShift MCP backends, OpenShift AI MCP catalog and gateway, Kuadrant, the registry; SDG Hub and OSFT for 403 tuning	[WG]
AT&T	OTEL 2.0 as the 403 model track; domain review of the skill catalog; the telecom tool-calling data, jointly	[WG]
Microsoft Azure	Cloud serving and evaluation runs at scale	[WG]
AMD	MI300X / MI355X serving of OTEL 2.0 at TP=1 for the evaluation track	[WG]
CNF vendors (onboarding)	MCP backends for the telco app layer: a capability manifest plus one offline conformance episode	[WG]

Decisions for the working group

DECIDED

4xx is the platform layer: each course builds one cross-workload component and proves it on the shipped courses without modifying them · the shipped courses stay as they are · new courses use skill and capability as defined; older courses keep their wording, with a glossary · the 401 to 404 ballot aliases are retired · CNF vendors stay unnamed in public until onboarding completes

Open decision	Options	Starting position
1 · Skill format	The open SKILL.md layout (harness-portable), or a repo-native schema	Open layout plus a small frontmatter extension: role, risk class, capabilities, offline episode
2 · Learner persona	Field architects, customer engineers, or the Purdue and LF Networking community; sets objective depth and exercise difficulty	WG to set
3 · OTEL serving home	AMD MI300X / MI355X at TP=1, Azure GPU, Venice's 96 GB card with a short context, or a MaaS endpoint	WG to set by what each party brings
4 · Vendor onboarding contract	A capability manifest plus an offline conformance episode (a tape with a golden trace), or live vendor lab access	Manifest plus episode is the contract; lab access optional

Step 1, the design note, starts after this session with decisions 1 and 4 settled; 2 and 3 can follow by the next meeting.